

Chargeback Prevention: The CFO Case

Cinderhaven Data Platform — Executive Summary

Executive Summary

Cinderhaven’s chargebacks are not inevitable. Most trace to a handful of upstream data conditions — and those conditions are fixable before the order ships.

\$446K

Chargebacks over 37 months

\$325K

Preventable at source

4 causes

Fixable root conditions

Grouping Cinderhaven’s historical chargebacks by root cause and applying a fixed preventability fraction to each category flags **73%** of the 37-month total (Jan 2023 – Jan 2026) as traceable to four upstream data conditions — missing case dimensions, barcode registration gaps, late ASN filing, and pricing discrepancies. This share is an assumption-driven estimate from the roadmap below, not a model output. Each condition is detectable before the order ships. Each has a specific operational fix.

The deductions team currently works these chargebacks reactively, winning 30–45% in disputes. Prevention requires no dispute process. Fixing the data conditions eliminates the chargeback at source and recovers the full amount, not a fraction of it.

Prevention roadmap

Table 1

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#	Root cause	Loss (37-mo)	Prevention value (37-mo)	Fix
1	Logistics Overage	\$245K	\$171K	Audit pick-and-pack procedures; add case
2	Data Compliance Error	\$143K	\$114K	Populate missing barcodes (GTIN-14, UP

#	Root cause	Loss (37-mo)	Prevention value (37-mo)	Fix
3	ASN Timing Infraction	\$42K	\$29K	Submit Advance Ship Notices within the
4	Pricing Discrepancy	\$17K	\$10K	Reconcile PO unit prices against approved

The economics

Dispute recovery returns 30–45 cents per dollar. Prevention returns 100 cents per dollar — and eliminates the working capital drag of post-ship disputes, the labor cost of the deductions team, and the carrier relationship risk from repeated chargebacks.

A companion model demonstrates the pre-ship scoring workflow. One caveat is essential to state plainly: on Cinderhaven’s *real* chargeback records the upstream features carry no predictive signal (AUC = 0.50 — real chargebacks are effectively random with respect to data quality). The model is therefore trained on **synthetic** labels drawn from a known formula (risk = ASN lateness $\times$ data-quality gaps), and it recovers that formula. On held-out synthetic data it flags 72% of chargebacks at a 0.5 threshold (recall) and reaches an AUC of 0.70. Predictive skill on the real target is effectively zero; the durable value here is the prevention roadmap, not the score. The question is which data conditions to fix first. This roadmap answers that by prevention value.

Produced using Cinderhaven synthetic demonstration data. Figures are illustrative of a repeatable methodology, not actual Cinderhaven financials. Real chargebacks carry no predictive signal (AUC 0.50); the model is trained on synthetic labels and reaches AUC 0.70 on held-out synthetic data.